

Omar Garib

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Research Interests

Intelligent systems for safety-critical autonomy; multimodal perception & interaction (speech/vision/3D spatial reasoning); human-in-the-loop decision support; robotics & human–robot teaming.

Education

Georgia Institute of Technology — Atlanta, GA

M.S. in Computational Science & Engineering | GPA: 4.0

Aug. 2024–Current

- Aerospace Systems Design Lab (ASDL), Digital Engineering (AI/ML) Division.
- **Focus:** intelligent-AI systems; automatic speech recognition (ASR); multimodal perception; LLM-based automated document extraction.

B.S. in Aerospace Engineering

Aug. 2020–May 2024

Research Experience

Aerospace Systems Design Lab (ASDL) — Atlanta, GA

Aug. 2024–Current

Graduate Research Assistant (Advisors: O. J. Pinon Fischer, D. N. Mavris)

Human-in-the-Loop Multimodal AI Simulation Environment for Aviation Conflict Detection:

- Developed an open simulation testbed for multimodal pilot/air-traffic-controller assistant systems with lightweight adapters for interchangeable model architectures/choices across ASR, vision, decision, and text-to-speech (TTS).
- Released a scripted scenario suite (terminal and en route conflicts) and a reference end-to-end pipeline; enabled rapid comparison and iteration of intelligent-system architectures across hundreds of reproducible runs.

GAN-Based Radio-Noise Augmentation and Fine-Tuned ASR for Air Traffic Control:

- Modeled air-traffic-control radio noise with a GAN-based pipeline that captures characteristic VHF artifacts and injected the learned noise during Whisper fine-tuning, yielding a word-error-rate (WER) drop from 14.66% to 3.58% on real ATC audio (75.6% relative reduction).

Multimodal Document Extraction for Simulation Repositories:

- Automated document/figure extraction and tagging via optical character recognition (OCR) + vision-language model (VLM) embeddings, and retrieval-augmented generation (RAG) with a large language model (LLM); improved discoverability of engineering artifacts.

NASA Langley Research Center (LaRC) — Hampton, VA (remote)

May 2025–Current

Researcher (Digital Transformation Initiative)

- Built ML pipelines and interactive visual analytics to accelerate engineering decision-making (further details restricted due to project confidentiality).
- Designed an automated document-extraction pipeline for engineering PDFs to facilitate integration into pre-existing NASA systems/models.

Industry Experience

Tesla — Palo Alto, CA

Aug. 2023–Dec. 2023

Mechanical Design Engineer Intern — Battery Structures

- Owned the design of 15+ battery-pack structural components; applied CAD & GD&T toward NHTSA collision-safety goals.
- Conducted 20+ FEA studies (Ansys/SimSolid); delivered a lifting system that improved test/manufacturing maneuverability and cut cost by ~35%.

Research Publications & Submissions

- **O. Garib**, J. D. Kambhampaty, O. J. Pinon Fischer, D. N. Mavris. “AIRHILT: A Human-in-the-Loop Testbed for Multimodal Conflict Detection in Aviation.” *Major robotics/ML conference (double-anonymous review)*. in submission **[Project]**.
- **O. Garib**, M. Ghanem, O. J. Pinon Fischer, D. N. Mavris. “SimuGAN-Whisper-ATC: Generative Noise Injection for Improved Automatic Speech Recognition in Air Traffic Control.” *AIAA SciTech*. accepted abstract **[PDF]**.
- **O. Garib***, M. M. Davis*, A. J. Kennedy, J. D. Kambhampaty, O. J. Pinon Fischer, D. N. Mavris. “Accelerating Knowledge Discovery in Engineering Simulation Repositories through Multimodal, LLM-Driven Metadata Tagging.” *AIAA SciTech*. accepted abstract **[PDF]**.

Coursework Projects

Dynamic Rollout in Model-Based Policy Optimization (MBPO) — Deep Reinforcement Learning

- Designed adaptive rollout schedulers (uncertainty-aware truncation, error/stability gating, gradient-based horizons) achieving ~25% higher return vs. fixed schedules on Hopper-v5 while avoiding long-horizon collapse.

Semantic Segmentation with PSPNet and Transfer — Computer Vision

- Trained and adapted a Pyramid Scene Parsing Network (PSPNet) for CamVid and KITTI road segmentation, achieving ~0.86 mIoU.

Visual Geometry & Panorama Stitching (from scratch) — Computer Vision

- Estimated projection and fundamental matrices; applied Random Sample Consensus (RANSAC); built custom Direct Linear Transform (DLT)/singular value decomposition (SVD) homography estimation with manual warping/blending.

LoRA-Fine-Tuned LLM for Dialog — Natural Language Processing

- Fine-tuned 4-bit OPT-1.3B with parameter-efficient fine-tuning/Low-Rank Adaptation (PEFT/LoRA); integrated top-p and beam search.

Technical Skills

ML/AI: Python, PyTorch, TensorFlow, MATLAB, Whisper (ASR), YOLO, OpenCV, FastAPI, REST/JSON, LLM/RAG, Vector DBs.